

Neural Network Architectures, a Reference

Match the architecture to the structure of the data.

Architecture	Common data types	Data structure	Inductive bias	Signature mechanism	Flagship example	Where it struggles
MLP (Day 1)	Tabular records, fixed feature vectors	A flat, unordered vector	None. Every input connects to every unit	Dense fully connected layers. A universal approximator	The Day 1 baseline for any vector task	Ignores structure and explodes in parameters on images
CNN	Images, video, audio spectrograms	Grids (a regular lattice of pixels or cells)	Locality and translation invariance	Convolution and pooling, a feature hierarchy from edges to objects, residual connections	Diabetic retinopathy screening; ImageNet	Not for graphs or for long-range order in sequences
RNN / LSTM	Text, speech, time series	Ordered sequences of variable length	Order and memory across time	A recurrent loop carrying a hidden state; the LSTM adds a cell state with forget, input, and output gates	Google Translate (2016); streamflow forecasting	Processes one step at a time, so slow and strained on very long ranges
GNN	Molecules, social and road networks, knowledge graphs, meshes	Graphs (nodes joined by edges, no fixed order)	Relations and permutation invariance	Message passing that aggregates neighbors by sum or mean, then a readout for the whole graph	GraphCast weather; halicin antibiotic	Over-smoothing if too deep; costly on very large graphs
Transformer (Day 4)	Text, images as patches, multimodal	Sets and sequences, with relations learned	Minimal hardwired structure; learns the relations through attention	Self-attention, positional encodings, residual connections and feed-forward blocks	BERT and GPT (tomorrow)	Data-hungry and compute-hungry, since little structure is built in

The spine: an MLP assumes no structure, the CNN, RNN, and GNN each assume one kind of structure, and the Transformer learns the structure itself.